

## Weekly Lesson Plan – Week 8 · Semester 1 · 2026–27 · Da Vinci School

| Teacher: Gavaldon   Week of: September 28, 2026   Course: Engineering Design & Problem Solving (InSPIRESS)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| <b>TEKS / ELPS:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                      |
| TEKS: §127.406(6) implement a design process to develop a multi-system product or solution<br>ELPS: c.1C, c.2C, c.3E, c.4C, c.4F, c.5G                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Aim of the Lesson / Objective:</b> <i>Statement of what you want students to do (skill) and/or know (knowledge). Students will...</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                      |
| MONDAY<br>SEP 28                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | TUESDAY<br>SEP 29                                                                                                                                                                                                                                                                                                                                                        | WEDNESDAY<br>SEP 30                                                                                                                                                                                                                                                                                                                                                            | THURSDAY<br>OCT 1                                                                                                                                                                                                                                                                                                                                  | FRIDAY<br>OCT 2                                                                                                                                                                                                                                                                                                                                                                                                      |
| Students will insert and calibrate a reference image and sketch a bottle profile using lines and a fit point spline.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Students will revolve a sketch profile into a solid and apply shell and fillet features to the resulting body.                                                                                                                                                                                                                                                           | Students will finish the bottle with an appearance and evaluate their own model against a checklist before submitting it.                                                                                                                                                                                                                                                      | Students will sketch a connected path and a profile and use the sweep feature to model a paperclip.                                                                                                                                                                                                                                                | Students will complete the paperclip model and compare extrude, revolve and sweep in their own words.                                                                                                                                                                                                                                                                                                                |
| <b>Employed Language Domain(s) Objective:</b> <i>How students achieve the objective – Listening, Speaking, Reading, Writing. Students will...</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Students will collaborate in teams to investigate engineering problems, communicate their findings, and present their solutions. They will listen to team members' ideas, discuss potential solutions, read relevant technical materials, and write explanations of their designs and findings.                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Assessment of Content Objective:</b> <i>Instrument used to determine mastery. Ex: Exit Ticket.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Students will be assessed by written, group, or oral responses throughout the class period.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Set (Bell Work):</b> <i>A motivating action that engages students in the objective – question, fact, activity. Essential Question:</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                      |
| MONDAY<br>SEP 28                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | TUESDAY<br>SEP 29                                                                                                                                                                                                                                                                                                                                                        | WEDNESDAY<br>SEP 30                                                                                                                                                                                                                                                                                                                                                            | THURSDAY<br>OCT 1                                                                                                                                                                                                                                                                                                                                  | FRIDAY<br>OCT 2                                                                                                                                                                                                                                                                                                                                                                                                      |
| Look at a plastic water bottle. Why is the outside shaped with ridges and curves instead of being a smooth cylinder?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | You find a mistake in a design your team already sent to the customer. Nobody else has noticed. Do you say something?                                                                                                                                                                                                                                                    | Why do bottles have a neck that is narrower than the body? Give two reasons.                                                                                                                                                                                                                                                                                                   | A paperclip is a single bent wire and it has not changed in over 100 years. Why has nobody improved it?                                                                                                                                                                                                                                            | You can make a product last twice as long for fifty cents more per unit, but then people buy half as many. What do you do?                                                                                                                                                                                                                                                                                           |
| <b>Methodology / Steps:</b> <i>What it takes to teach the objective so 100% can pass the assessment. Teacher will... Teams will... Individuals will...</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| Step 1: Bell ringer / essential question.<br>Step 2: Bell work: why is a plastic water bottle shaped with ridges instead of being a smooth cylinder? (10 min)<br>Step 3: Save the bottle picture; new file saved in your folder (5 min)<br>Step 4: Day 2 video: Insert > Canvas, calibrate, sketch half the outline (28 min)<br>Step 5: Save (Ctrl+S) before the bell (2 min)                                                                                                                                                                                                                                                                                             | Step 1: Bell ringer / essential question.<br>Step 2: Bell work (dilemma): you find a mistake in a design already sent to the customer - do you say something? (10 min)<br>Step 3: Day 2 video: revolve around the center line, shell through the top, fillet the edges (30 min)<br>Step 4: Save (Ctrl+S) (3 min)<br>Step 5: Check: is it round from every angle? (2 min) | Step 1: Bell ringer / essential question.<br>Step 2: Bell work: why is a bottle neck narrower than the body? Give two reasons. (10 min)<br>Step 3: Finish the bottle: anything left, then drag on the glass appearance (27 min)<br>Step 4: Self-check against the list; tell me if it is not going to be done (6 min)<br>Step 5: Save (Ctrl+S) in Students > your name (2 min) | Step 1: Bell ringer / essential question.<br>Step 2: Bell work: a paperclip has not changed in 100 years - why has nobody improved it? (10 min)<br>Step 3: New file: Day 3 - Paperclip - Lastname, units in mm (5 min)<br>Step 4: Day 3 video: sketch the path, sketch the circle, sweep (28 min)<br>Step 5: Save (Ctrl+S) before the bell (2 min) | Step 1: Bell ringer / essential question.<br>Step 2: Bell work (dilemma): make it last twice as long for 50 cents more, and sell half as many? (10 min)<br>Step 3: Finish the sweep; adjust the wire thickness; save (22 min)<br>Step 4: Folder check: block, bottle, paperclip - all three in Students > your name (8 min)<br>Step 5: Three tools, three builds: write what each one does in your own words (5 min) |
| <b>Resources:</b> <i>Textbook, handouts, visuals, laptops.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Google Classroom, projector, engineering notebook, Design Brief packet / worksheets, laptops or phones for AI tools.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Re-Teaching Activities:</b> <i>What you prepare if students do not master the objective.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Review during bellwork; small-group re-teach at the teacher table; one-on-one explanation on request; peer and native-language support within the seating group.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Reflection:</b> <i>Students analyze what they accomplished and how they reached mastery.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Exit ticket – students state what they accomplished and what worked best.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Study Skills:</b> <i>Skills implemented by students – note-taking, summarizing, teamwork, etc.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Note-taking, summarizing, outlining, reading comprehension, teamwork, oral communication, problem-solving.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>SPED &amp; 504 Accommodations / Modifications:</b> <i>Codes.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Per Individualized Education Plan (IEP) / 504 plan.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>ELL Accommodations:</b> <i>Codes.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Clarify directions (CD); Extra time (ET); Peer and Native Language Support (PN); Pre-teach vocabulary (PV); Rephrase, Repeat, Slow Down (RS); Visual Cues (VC); Wait time (WT); Pictorial Representation (DP). GROUPING: students are seated in mixed-proficiency small groups so emergent bilingual students hear and use the academic vocabulary in context, rehearse with a peer before whole-class response, and can be pulled as a small group for re-teach (PN, WT). ONE-ON-ONE: individual teacher explanation is available on request every period; directions are restated and modeled individually at the student desk after whole-class delivery (CD, RS, VC). |                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                      |